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AnomalyNCD: Towards Novel Anomaly Class Discovery in Industrial Scenarios

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Abstract

Recently, multi-class anomaly classification has garnered increasing attention. Previous methods directly cluster anomalies but often struggle due to the lack of anomaly-prior knowledge. Acquiring this knowledge faces two issues: the non-prominent and weak-semantics anomalies. In this paper, we propose AnomalyNCD, a multi-class anomaly classification network compatible with different anomaly detection methods. To address the non-prominence of anomalies, we design main element binarization (MEBin) to obtain anomaly-centered images, ensuring anomalies are learned while avoiding the impact of incorrect detections. Next, to learn anomalies with weak semantics, we design mask-guided representation learning, which focuses on isolated anomalies guided by masks and reduces confusion from erroneous inputs through corrected pseudo labels. Finally, to enable flexible classification at both region and image levels, we develop a region merging strategy that determines the overall image category based on the classified anomaly regions. Our method outperforms the state-of-the-art works on the MVTec AD and MTD datasets. Compared with the current methods, AnomalyNCD combined with zero-shot anomaly detection method achieves a 10.8% F_1 gain, 8.8% NMI gain, and 9.5% ARI gain on MVTec AD, and 12.8% F_1 gain, 5.7% NMI gain, and 10.8% ARI gain on MTD. Code is available at https://github.com/HUST-SLOW/AnomalyNCD.

1. Introduction

In recent years, industrial anomaly detection [22, 24, 31, 53] has garnered remarkable performance. It locates visual anomalies on industrial products but without recognizing fine-grained anomaly categories. For downstream anomaly treatments, it is necessary to recognize the anomaly classes, such as fracture, ablation, etc., and even discover novel anomaly categories that are constantly emerging.

Several anomaly clustering approaches [1, 28, 40] have been proposed to address the multi-class anomaly classification task, as illustrated in Fig. 1 (a). These approaches typically employ a two-step process: anomaly localization followed by feature extraction from the anomalous regions for clustering. However, the effectiveness of these methods is limited when dealing with homogeneous anomaly patterns that exhibit differences in shape, appearance, and location, primarily due to the lack of prior knowledge of anomaly types. Therefore, our study aims to classify unseen anomalies by leveraging prior knowledge of seen anomalies, thereby overcoming the limitations of current methods.

Through detailed investigation, we find two primary obstacles in leveraging prior knowledge of known anomalies, which prevent existing classification network to industrial anomalies. ❶ **Non-prominence Anomalies**: Classification in natural scenes assumes that subjects are centered in the images so that networks easily extract their semantics, while this assumption is not true in industrial scenarios. ❷ **Low-semantics Anomalies**: In contrast to natural objects, industrial anomalies are typically less semantic, making the network hard to focus on anomalies but prone to background.

In this paper, we overcome both obstructions and propose a multi-class anomaly classification network that aligns with the concept of novel class discovery (NCD), as illustrated in 1 (c). This network learns to classify isolated anomalies by focusing the model's attention on real anomaly regions. Firstly, to address the non-prominence issue, we propose a main element binarization (MEBin) method. MEBin isolates the primary anomaly regions by extracting them from anomaly detection (AD) results. This binarization process alleviates the negative impact of errors within AD results on the subsequent multi-class anomaly learning, enabling compatibility with various AD algorithms. Secondly, to tackle low-semantics anomalies, we propose mask-guided representation learning. It leverages the anomaly masks generated by MEBin to direct the network's attention specifically to individual anomalous re-

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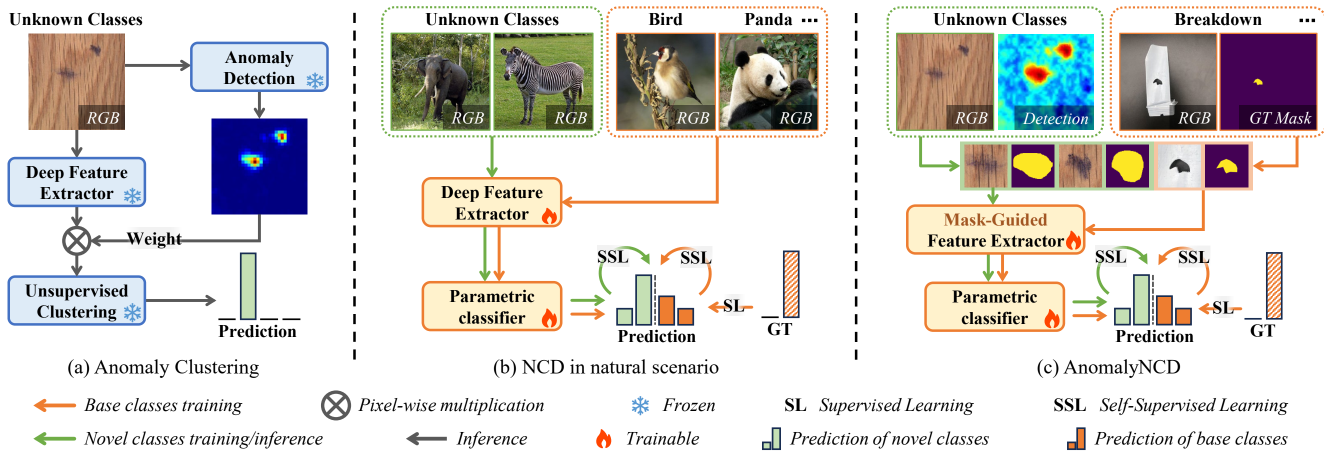

Figure 1. **Comparison between solutions organizing anomalies into groups.** (a) Anomaly clustering methods extract the features of the anomaly region and employ unsupervised clustering algorithms to cluster the anomalies. (b) Vanilla NCD methods typically employ a trainable feature extractor and classifier. These components process object-centered images from both known and unknown classes. (c) Our method aims to learn features of isolated anomaly regions using anomaly-centered sub-images and masks from MEBin.

gions, ensuring our network learns more discriminative features specific to anomalies and classifies each individual anomaly. Additionally, we use corrected pseudo labels in this learning method to prevent false-positive anomalies from training. Finally, to achieve more flexible classification in both region and image levels, we propose a region merging strategy, which determines the image category according to the classification of each region in the image. Experimental results on the MVTec AD [5] and MTD datasets [23] demonstrate the effectiveness of AnomalyNCD.

In summary, the major contributions of our work are:

- We propose the first method based on self-supervised manner in multi-class anomaly classification, named AnomalyNCD. It can be seamlessly combined with different anomaly detection methods and classify unseen anomalies detected into homogeneous groups.
- We study the challenges of learning from anomalous regions in industrial scenarios, motivating us to propose MEBin and mask-guided representation learning for our network to focus on real anomalous regions and learn discriminative features of isolated anomalies.
- AnomalyNCD surpasses existing anomaly clustering methods and NCD methods on the MVTec AD and MTD datasets, providing a decent foundation for downstream anomaly treatments in industrial scenarios.

2. Related Work

2.1. Image Clustering in Industrial Scenarios

Directly clustering the images cannot perform well in multi-class anomaly classification. To address this issue, two studies build the clustering pipeline specific for fine-grained anomaly classes. Sohn *et al.* [40] proposed a simple but effective method to aggregate the patch features by anomaly map weighting. Lee *et al.* [28] only aggregates the patch features with the higher anomaly score to determine the classes. However, both methods rely on frozen models that cannot learn features specific to anomalies. Therefore, we adopt self-supervised learning to discover novel classes from unlabeled anomalies.

2.2. Novel Class Discovery

Novel Class Discovery setting was first formalized in [18]. The early methods follow a two-stage pipeline [18, 20, 21]. The first stage is to learn representation from the base set to obtain prior knowledge for classification, which serves as the basis for the second one to train the model on the novel set. Recent methods are one-stage [19, 46, 56]. By handling the base set and novel set simultaneously, one-stage methods extract better representations with less bias towards the base classes. NCD has been applied in several fields, such as medical image classification [14, 58], semantic segmentation [55] and point cloud segmentation [37, 49]. In this paper, we introduce the NCD architecture into multi-class industrial anomaly classification for the first time.

2.3. Binarization Approach

Binarization is a crucial preprocessing step for various vision tasks, e.g., edge detection [6, 59], segmentation [10, 16, 57], document image processing [7, 34, 41], medical image processing [39, 42], object detection [50–52], etc. Otsu [35] determines the optimal threshold to separate the foreground and background by maximizing the between-class variance of the pixel intensities. DiffuMask [48] utilizes cross-attention maps and adaptive thresholding to generate binary maps. However, these methods usually obtain fragmented false-positive regions when segmenting the anomalies, which harms the region-level anomaly class prediction. To alleviate this, we propose the main element

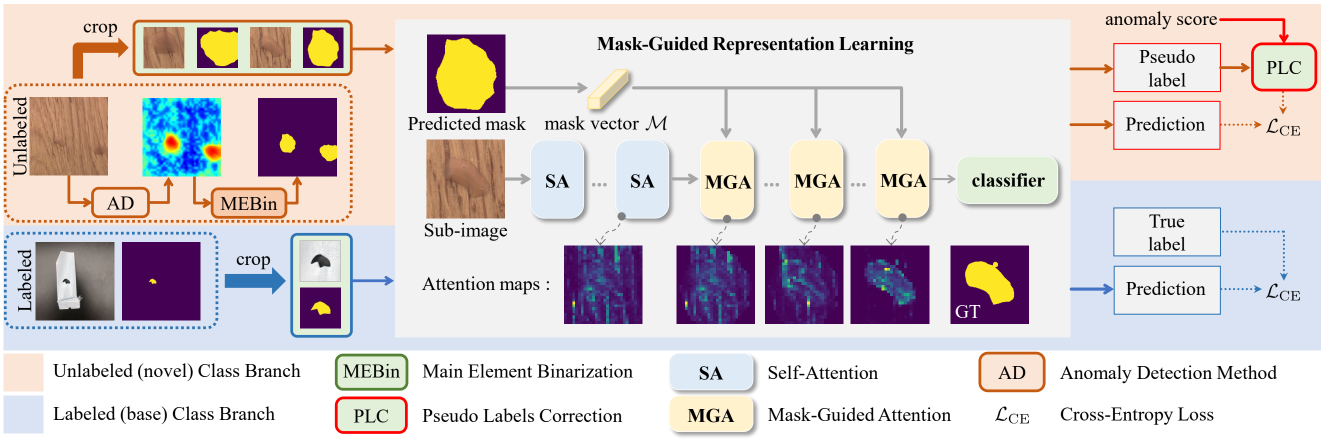

Figure 2. **Illustration of the Training Process for AnomalyNCD**. First, we apply main element binarization (MEBin) to segment anomaly masks from detection results and generate anomaly-centered sub-images. Second, we introduce mask-guided representation learning to learn discriminative features of anomalies, classifying sub-images into various categories.

binarization strategy to extract only the primary anomaly region, thereby reducing the impact of false positives on representation learning.

3. Method

The proposed AnomalyNCD aims to automatically discover and recognize the visual categories of industrial anomalies. Fig. 2 illustrates its overall framework. To extract main anomaly regions from detection results, we outline the main element binarization method (Sec. 3.2). Guided by these anomaly regions, we then apply mask-guided representation learning (Sec. 3.3) to extract and classify discriminative features of anomalies within these regions. Finally, a region merging strategy (Sec. 3.4) ensures robust image-level classification based on all anomaly regions.

3.1. Problem Definition

Given a set of unlabelled images $\mathcal{D}^{\mathbf{u}} = \{I_i^{\mathbf{u}} | i \in [1, N^{\mathbf{u}}]\}$, the goal of AnomalyNCD is to assign them into $\mathcal{C}^{\mathbf{u}}$ classes (named "**Novel**" previously), where $\mathcal{C}^{\mathbf{u}}$ is known as a priori, referring to vanilla NCD [15, 17, 45]. To provide a better foundation for learning novel classes, we follow previous NCD tasks to assume a set of labeled anomaly images $\mathcal{D}^{\mathbf{l}} = \{(I_i^{\mathbf{l}}, y_i^{\mathbf{l}}, \mathbf{M}_i^{\mathbf{l}}) | i \in [1, N^{\mathbf{l}}]\}$ with $\mathcal{C}^{\mathbf{l}}$ classes available (named "**Base**" previously), where $y_i^{\mathbf{l}} \in \mathbb{R}^{1\times(\mathcal{C}^{\mathbf{l}}+\mathcal{C}^{\mathbf{u}})}$ is the one-hot class label, and $\mathbf{M}_i^{\mathbf{l}}$ is the ground truth anomaly mask of image $I_i^{\mathbf{l}} \in \mathcal{D}^{\mathbf{l}}$. $\mathcal{D}^{\mathbf{l}}$ supplies the auxiliary knowledge of the industrial anomalies for grouping the images in $\mathcal{D}^{\mathbf{u}}$. To further generalize our study, we also perform experiments with different $\mathcal{D}^{\mathbf{l}}$ and without $\mathcal{D}^{\mathbf{l}}$ in the Appendix F.

3.2. Main Element Binarization

To provide a clear anomaly indication of unlabeled images, we use an anomaly detection method on $I_i^{\mathbf{u}}$ to generate the anomaly probability map, denoted as $A_i \in [0, 1]^{H\times W}$. However, A_i inevitably contains false positives (over-detections) and false negatives (missed detections). These errors can negatively affect multi-class anomaly classification. To address this, we propose a Main Element Binarization (MEBin) approach. It extracts the principal anomaly structures (Main Element) from A_i by focusing on stable regions across small threshold variations. Overall, the MEBin consists of three steps: ❶ Predefine a range $[\mathbf{s}_{\min}, \mathbf{s}_{\max}]$ for threshold exploration; ❷ Binarize A_i across all thresholds within this range; ❸ Identify the main elements in A_i based on the changes of segmented regions under different binarization results.

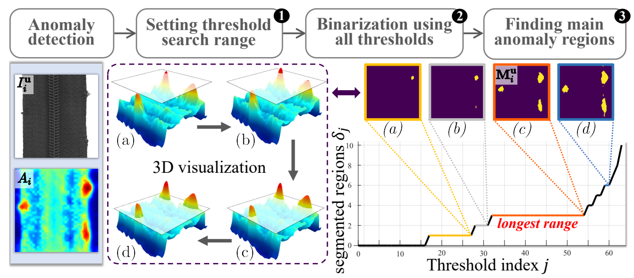

Figure 3. **Pipeline of the Main Element Binarization.** In 3D visualization, we show the changes of segmented regions under different thresholds. The 3D map (a) corresponds to the 2D binary segmentation mask *(a)*.

The first step: To capture all potential anomalies across the set $\{A_i | i \in [1, N^{\mathbf{u}}]\}$, the upper limit $\mathbf{s}_{\max}$ is set to 1. Conversely, the lower limit, $\mathbf{s}_{\min}$, is determined by the minimum anomaly score[1] found across the set $\{A_i | i \in [1, N^{\mathbf{u}}]\}$, represented as $\mathbf{s}_{\min} = \min(s_1, s_2, ..., s_{N^{\mathbf{u}}})$, where $s_i = \max(A_i)$. Such a lower limit helps minimize the likelihood of normal image pixels being segmented.

The second step: To analyze the segmented regions

[1] Anomaly score: Maximum value in an anomaly probability map.

change with thresholds, we uniformly sample $\mathcal{T}$ thresholds $\{\epsilon_j | j \in [1, \mathcal{T}]\}$ in $[\mathbf{s}_{\min}, \mathbf{s}_{\max}]$. For each threshold ϵ_j, the anomaly map A_i is binarized into a mask M_i^j: $M_i^j = \mathbb{1}[A_i > \epsilon_j]$, where $\mathbb{1}[.]$ is the indicator function. To reduce the meaningless and fragmented segmentation, we further apply an erosion operation to each mask M_i^j.

The third step: We identify main anomaly regions through two operations. First, let δ_i^j denote the number of individual regions segmented in M_i^j, each region is an independent connected component. We observe that the amount of the main regions in the anomaly map A_i can be characterized by the non-zero value that appears most frequently in $\{\delta_i^j | j \in [1, \mathcal{T}]\}$, denoted as $\bar{\delta}_i$. Second, to segment the main regions, we take the longest continuous threshold range where only $\bar{\delta}_i$ regions are segmented. The main anomaly regions remain stable within this threshold range, as shown in Fig. 3.

- If this range is shorter than τ, there is no anomaly region to be segmented.
- Conversely, if this range exceeds τ, it confirms stable anomaly segmentation within this threshold range. In this case, we select the minimum threshold from the identified range to segment anomalies as completely as possible.

Finally, we use the selected threshold to segment A_i into the binary mask, denoted as $\mathbf{M}_i^{\mathbf{u}}$. In this way, the unlabeled images set $\mathcal{D}^{\mathbf{u}}$ are extended with the binary masks: $\{(I_i^{\mathbf{u}}, \mathbf{M}_i^{\mathbf{u}}) | i \in [1, N^{\mathbf{u}}]\}$.

Anomaly-Centered Sub-Image Cropping. To focus the model on anomalous regions, we crop each region in $\mathbf{M}_i^{\mathbf{u}}$ individually into anomaly-centered sub-images. Specifically, for each region in $\mathbf{M}_i^{\mathbf{u}}$, we use the minimum bounding square to crop the sub-image from $I_i^{\mathbf{u}}$ and its mask from $\mathbf{M}_i^{\mathbf{u}}$. Then the sub-images of $I_i^{\mathbf{u}}$ can be denoted as $\{x_{i,1}^{\mathbf{u}}, x_{i,2}^{\mathbf{u}}, \ldots\}$, and the corresponding masks cropped from $\mathbf{M}_i^{\mathbf{u}}$ are denoted as $\{m_{i,1}^{\mathbf{u}}, m_{i,2}^{\mathbf{u}}, \ldots\}$. To align labeled and unlabeled inputs, we use the same cropping operation to extract sub-images and masks for the labeled data set $\mathcal{D}^{\mathbf{l}}$, denoted as $\{(x_{i,1}^{\mathbf{l}}, m_{i,1}^{\mathbf{l}}), (x_{i,2}^{\mathbf{l}}, m_{i,2}^{\mathbf{l}}), \ldots | i \in [1, N^{\mathbf{l}}]\}$. These are then merged with the unlabeled data in the same structures, represented as $\{(x_{i,k}, m_{i,k}, y_{i,k}) | i \in [1, N^{\mathbf{l}} + N^{\mathbf{u}}], k \in [1, \bar{\delta}_i]\}$, where $\bar{\delta}_i$ indicates the number of sub-images per image, and $y_{i,k}$ is the one-hot class label for each labeled sub-image, matching the label of the i^{th} image if it belongs to $\mathcal{D}^{\mathbf{l}}$. Implementation details of the cropping process are supplied in the Appendix C.

3.3. Mask-Guided Representation Learning

3.3.1. Mask-Guided Vision Transformer (MGViT)

MGViT is built on vision transformer (ViT). Thus we review its structure.

Vision transformer [13] processes an input image by dividing it into $\mathtt{N}$ fixed-size patches. Each patch is treated as a "token" and is flattened into an embedding vector of length D. ViT has L sequential transformer layers, each containing self-attention layers and feed-forward neural networks. In addition, a special class token `[CLS]`, is propagated from the first layer to the final layer where it serves as the basis for classification. In the self-attention of the l^{th} layer, the input patches undergo linear projections to form the queries, keys, and values $\mathbf{Q}_{l-1}, \mathbf{K}_{l-1}, \mathbf{V}_{l-1} \in \mathbb{R}^{(\mathtt{N}+1) \times D}$. Here, ViT captures the relationships between the image patches by self-attention mechanism $Attn$ as follows:

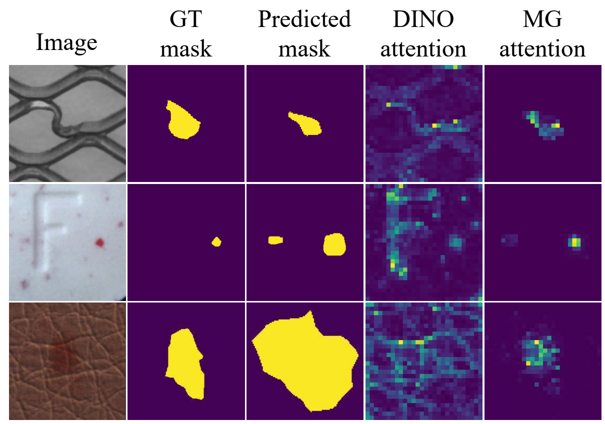

Figure 4. **Visualization of the self-attention of the `[CLS]` token on the last layer's heads.** DINO attention refers to the `[CLS]` token extracted from a DINO pre-trained ViT that mainly focuses on a foreground object. AnomalyNCD uses a mask to direct the `[CLS]` token's attention to the anomalous regions.

$$Attn = \text{softmax}\big(\text{concat}(\mathbf{Q}_{l-1}^{\text{cls}} \mathbf{K}_{l-1}^{\top}, \mathbf{Q}_{l-1}^{\text{patch}} \mathbf{K}_{l-1}^{\top})\big) \mathbf{V}_{l-1} \tag{1}$$

where we divide $\mathbf{Q}_{l-1}$ into $\mathbf{Q}_{l-1}^{\text{cls}} \in \mathbb{R}^{1 \times D}$ and $\mathbf{Q}_{l-1}^{\text{patch}} \in \mathbb{R}^{\mathtt{N} \times D}$ to represent class token and patch tokens separately.

Insufficient Focus on Anomalies. The ViT excels in semantic representation but fails to detect weak semantic anomalies. Fig. 4 demonstrates this using the self-attention map of the `[CLS]` token in the last layer of a DINO-pretrained ViT [9]. In the 1st row's example, the model highlights the salient object (metal grid) in an image, rather than the local anomaly. This observation means that the network over-focuses on learning background.

Mask-Guided Attention. To address the issue above, we propose mask-guided attention to direct ViT's focus on the anomalous region. Specifically, the input sub-image $x_{i,k}$ is fed into the ViT, and we assume the token number is $\mathtt{N}$. At the same time, its mask $m_{i,k}$ is resized to $\sqrt{\mathtt{N}} \times \sqrt{\mathtt{N}}$ by average pooling and then flattened into a vector $\mathcal{M} \in \mathbb{R}^{\mathtt{N} \times 1}$. Subsequently, we insert the constant 1 at the beginning of $\mathcal{M}$ to align its size with the tokens. Next, we use the mask vector in the self-attention mechanism for re-directing focus, which has three possible design choices.

1. Adding the mask $\mathcal{M}$ on both class and patch tokens.
2. Adding the mask $\mathcal{M}$ on the patch tokens.
3. Adding the mask $\mathcal{M}$ on the class token.

The 1st and 2nd designs suppress the rich contextual features in patch tokens that are crucial for classification, and our network only focuses on classifying anomalies rather than segmenting. Thus, the 3rd one is recommended as the most suitable design, and the experiment also verifies its effectiveness. Such a design can be formulated as follows:

$$Attn = \text{softmax}\left(\text{concat}(\mathbf{Q}^{\text{cls}}_{l-1}\mathbf{K}^{\top}_{l-1} + \overline{\mathcal{M}}\,, \mathbf{Q}^{\text{patch}}_{l-1}\mathbf{K}^{\top}_{l-1})\right)\mathbf{V}_{l-1} \tag{2}$$

where $\overline{\mathcal{M}}$ denotes the converted mask vector that is compatible with the attention mechanism [12]:

$$\overline{\mathcal{M}}(i) = \begin{cases} 0, & \text{if } \mathcal{M}(i) > 0.5 \\ -\infty, & \text{otherwise} \end{cases} \tag{3}$$

Note that, in the ViT, the self-attention in the last L_m-layer is replaced with the mask-guided attention to form MGViT. Fig. 4 reveals that the [CLS] token focuses on anomalous regions by using our mask-guided attention. Even though the predicted masks do not align with the ground truth accurately, the network still focuses on the anomaly roughly.

3.3.2. Model Training

Given a sub-image pair $(x_{i,k}, m_{i,k})$, we first apply random augmentation to generate two views $(\tilde{x}_{i,k}, \tilde{m}_{i,k})$ and $(\hat{x}_{i,k}, \hat{m}_{i,k})$. Following DINO [9], we feed these two views into the teacher and student networks respectively, which have shared MGViT and classification heads with different softmax temperatures. For labeled sub-images, the ground truth labels are converted into one-hot vectors to serve as supervision targets for the student network. For unlabeled sub-images $\tilde{x}_{i,k}, \hat{x}_{i,k}$, we generate pseudo labels $\hat{q}_{i,k}, \tilde{q}_{i,k} \in \mathbb{R}^{1\times(\mathcal{C}^{\mathbf{l}}+\mathcal{C}^{\mathbf{u}})}$ by the "teacher" one, which employs a sharp temperature τ_t to produce confident predictions. The entries of labeled classes ($\mathcal{C}^{\mathbf{l}}$) in the pseudo labels are set to 0. The "student" one, using a smooth temperature τ_s, is trained to align its probability predictions $\hat{p}_{i,k}$, $\tilde{p}_{i,k} \in \mathbb{R}^{1\times(\mathcal{C}^{\mathbf{l}}+\mathcal{C}^{\mathbf{u}})}$ with either ground truth labels (for labeled data) or pseudo labels (for unlabeled data). This dual supervision ensures effective learning from both labeled and unlabeled data.

Our training objective follows existing category discovery methods [45, 47] and includes classification objective and representation learning objective. The classification objective contains ground truth label supervision $\mathcal{L}^{\mathbf{l}}_{\text{cls}}$ on labeled sub-images, and pseudo label supervision $\mathcal{L}^{\mathbf{u}}_{\text{cls}}$ on unlabeled sub-images. The representation learning objective contains self-supervised contrastive learning $\mathcal{L}_{\text{rep}}$ [11] on both labeled sub-images and unlabeled sub-images, and supervised contrastive learning $\mathcal{L}^{\mathbf{l}}_{\text{rep}}$ [26] on only labeled sub-images. Finally, we adopt a mean-entropy maximization regularizer $\mathcal{L}^{\mathbf{u}}_{\text{reg}}$ [8] for unlabeled sub-images. More details about the training procedure are provided in Appendix A.

The overall loss of training objective is written as:

$$\mathcal{L} = \lambda(\mathcal{L}^{\mathbf{l}}_{\text{rep}} + \mathcal{L}^{\mathbf{l}}_{\text{cls}}) + (1-\lambda)(\mathcal{L}_{\text{rep}} + \mathcal{L}^{\mathbf{u}}_{\text{cls}} + \mu\mathcal{L}^{\mathbf{u}}_{\text{reg}}) \tag{4}$$

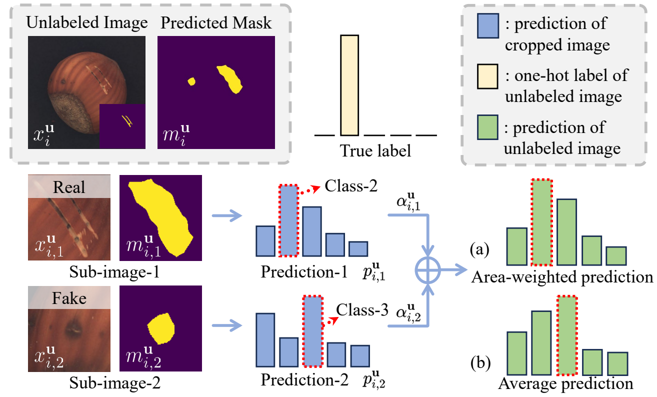

Figure 5. **Region merging strategy for image-level classification.** Using average prediction, the output of normal cropped images (Sub-image-2) leads to final misclassification, while area-weighted prediction can reduce the negative effect of normal cropped images on the final result.

where λ is a hyperparameter that balances the supervised loss and the self-supervised loss. μ is the coefficient of the regularization.

Pseudo Labels Correction. The error in pseudo labels leads to the false learning objective for the classification model. To mitigate these errors, especially the false detection of anomalous regions, we propose to use anomaly scores (described in Sec 3.2) to correct pseudo labels. Let $s_{i,k}$ denote the anomaly score for $x_{i,k}$, we weightedly judge whether the view $\hat{x}_{i,k}$ should be labeled as normal or assigned the generated pseudo labels.

$$\hat{q}_{i,k} \leftarrow w_{i,k}\mathbf{e} + (1-w_{i,k})\hat{q}_{i,k};\quad w_{i,k} = \max(0.5 - s_{i,k}, 0) \tag{5}$$

where $w_{i,k}$ is the weight for normal. $\mathbf{e} \in \mathbb{R}^{1\times(\mathcal{C}^{\mathbf{l}}+\mathcal{C}^{\mathbf{u}})}$ denotes a unit vector whose $(\mathcal{C}^{\mathbf{l}}+1)$-th value is 1 and all other values are 0, in which we use the first class in the $\mathcal{C}^{\mathbf{u}}$ classes as normal. If $s_{i,k}$ is smaller, the more likely the sub-image $\hat{x}_{i,k}$ is normal, and $\hat{q}_{i,k}$ is closer to $\mathbf{e}$. Thus, such correction forces the network to separate normal sub-images from the train set. In the same way, the pseudo label $\tilde{q}_{i,k}$ is updated.

3.4. Region Merging for Image Classification

For flexible classification in both sub-image and image levels, we aim to determine the image's anomaly class according to its sub-image classification. Intuitively, a naive method is to average the predictions of all sub-images within an image. However, false classification of the over-detected region may mislead the image class. As shown in Fig. 5, *Sub-image-1* and *Sub-image-2* are classified to *Class-2* and *Class-3*, respectively, where the wrong prediction of *Sub-image-2* misleads the image class in Fig. 5 (b).

To address this issue, we propose a region merging approach to robustly classify images. We found that misclassification is mostly over-detected regions that are much

Datasets	Metric	**IIC[25]**	**GATCluster[33]**	**SCAN[44]**	**UNO[15]**	**GCD[45]**	**SimGCD[47]**	**AMEND[3]**	**AC[40] (Unsup.)**	**MuSc [30] +AnomalyNCD**
MVTec AD [5]	NMI	0.093	0.136	0.210	0.146	0.417	0.452	0.431	<u>0.525</u>	**0.613**
	ARI	0.020	0.053	0.103	0.052	0.302	0.346	0.333	<u>0.431</u>	**0.526**
	F_1	0.285	0.264	0.335	0.342	0.553	0.569	0.542	<u>0.604</u>	**0.712**
MTD [23]	NMI	0.064	0.028	0.041	0.034	<u>0.211</u>	0.105	0.138	0.179	**0.268**
	ARI	0.020	0.009	0.029	0.011	0.115	0.048	0.067	<u>0.120</u>	**0.228**
	F_1	0.252	0.243	0.282	0.221	<u>0.381</u>	0.293	0.324	0.346	**0.509**

Table 1. Quantitative results on the MVTec AD and MTD dataset. All the methods only use unlabeled images as input. The best-performing result is in bold, the second best result is underlined.

Datasets	Metric	**AC[40] (Semi-sup.)**	**UniFormaly[28]** (0.953 / 0.837)	**PatchCore[38]** (0.938 / 0.729) **+AnomalyNCD**	**RD++[43]** (0.950 / 0.741) **+AnomalyNCD**	**EfficientAD[4]** (0.917 / 0.731) **+AnomalyNCD**	**PNI[2]** (0.942 / 0.516) **+AnomalyNCD**	**CPR[29]** (0.964 / -) **+AnomalyNCD**
MVTec AD [5]	NMI	0.608	0.547	0.670	0.631	0.516	<u>0.675</u>	**0.736**
	ARI	0.489	0.433	0.601	0.542	0.394	<u>0.609</u>	**0.674**
	F_1	0.652	0.645	<u>0.769</u>	0.721	0.641	<u>0.769</u>	**0.805**
MTD [23]	NMI	<u>0.390</u>	**0.421**	0.380	0.368	0.220	0.181	-
	ARI	0.314	0.322	**0.390**	<u>0.361</u>	0.188	0.219	-
	F_1	0.490	<u>0.609</u>	**0.617**	0.600	0.467	0.465	-

Table 2. Quantitative results on the MVTec AD and MTD dataset. All the methods use unlabeled images and labeled normal images as input. We evaluate the AUPRO metric across various anomaly detection approaches on two datasets, e.g., (0.953 / 0.837), the first one on the MVTec AD dataset and the second on the MTD dataset. For the metrics of multi-class anomaly classification, the best-performing result is in bold, the second best result is underlined. The unofficial results are marked in gray .

smaller than real anomalies. Therefore, we define an area-related weight $\alpha^{\mathbf{u}}_{i,k}$ for each sub-image $x^{\mathbf{u}}_{i,k}$, which is used in determining the image category:

$$\alpha^{\mathbf{u}}_{i,k} = \frac{\exp(\sqrt{\mathbf{a}^{\mathbf{u}}_{i,k}}/\tau_\alpha)}{\sum_{k=1}^{\bar{\delta}_i}\exp(\sqrt{\mathbf{a}^{\mathbf{u}}_{i,k}}/\tau_\alpha)} \tag{6}$$

where $\mathbf{a}^{\mathbf{u}}_{i,k}$ is the area of anomaly in $x^{\mathbf{u}}_{i,k}$, and τ_α is the temperature value. Then, for an image $I^{\mathbf{u}}_i$ composed by sub-images $\{x^{\mathbf{u}}_{i,k}|k \in [1, \bar{\delta}_i]\}$, the prediction logit is $p^{\mathbf{u}}_i = \sum_{k=1}^{\bar{\delta}_i} \alpha^{\mathbf{u}}_{i,k} p^{\mathbf{u}}_{i,k}$, where $p^{\mathbf{u}}_{i,k}$ is the probability prediction of $x^{\mathbf{u}}_{i,k}$ generated by the student network. As shown in Fig. 5 (a), the prediction $p^{\mathbf{u}}_i$ depends more on $p^{\mathbf{u}}_{i,1}$, which has a larger weight $\alpha^{\mathbf{u}}_{i,1}$ and a correct prediction.

4. Experiments

4.1. Experimental setting

4.1.1. Datasets

We conduct experiments on industrial datasets MVTec AD [5] and Magnetic Tile Defect (MTD) [23]. MVTec AD has industrial product images of 10 object categories and 5 texture categories. Each product category has at least two anomaly classes. Following [40] and [28], we remove the combined anomaly classes for a fair comparison. Notice that our method can also handle the combined anomaly class, as described in the Appendix G. MTD dataset has 952 normal images and 392 abnormal images, the abnormal images are divided into five anomaly classes. We follow [40] that uses 80% of normal images as the reference images and the rest as the test images. For both datasets, we use the single-blade sub-dataset in Aero-engine Blade Anomaly Detection Dataset (AeBAD-S) [54] with the normal class removed, as our default labeled image set $\mathcal{D}^{\mathbf{l}}$.

4.1.2. Implementation details

We use ViT-B/8 [13] pre-trained with DINO [9] as our feature extractor. The self-attentions in the last 9 layers are replaced with our mask-guided attention. During training, all the layers of ViT are fixed except the last layer. For anomaly detection methods preceding AnomalyNCD, we choose zero-shot method MuSc [30], one-class methods PatchCore [38], EfficientAD [4], RD++ [43], PNI [2] and CPR [29]. More details are in the Appendix B.

4.1.3. Competing methods

We compare our method with two state-of-the-art industrial anomaly clustering methods, UniFormaly [28] and Anomaly Clustering [40] that has two settings: an unsupervised one (Unsup.) and a semi-supervised one (Semi-sup.). The former setting only inputs unlabeled images for clustering, while the latter one uses labeled normal images, same as the input of one-class anomaly detection. We also employ three deep clustering methods, IIC [25], GATCluster [33], and SCAN [44], which cluster unlabeled images directly. In addition, four NCD methods in the nature scene are considered for comparison, UNO [15], GCD

	Avg FPR ↓	Avg FNR ↓	NMI ↑	ARI ↑	F_1 ↑
$\epsilon = 0.1$	0.572	0.617	0.554	0.482	0.650
$\epsilon = 0.3$	0.678	0.742	0.499	0.404	0.587
$\epsilon = 0.5$	0.484	0.165	0.567	0.458	0.640
$\epsilon = 0.7$	0.269	0.247	0.495	0.395	0.623
$\epsilon = 0.9$	0.544	0.593	0.077	0.013	0.337
Otsu [35]	0.676	0.525	0.382	0.268	0.499
Ours	**0.153**	**0.035**	**0.613**	**0.526**	**0.712**

Table 3. Ablation study of different binarization approaches on the MVTec AD dataset. The best-performing result is in bold.

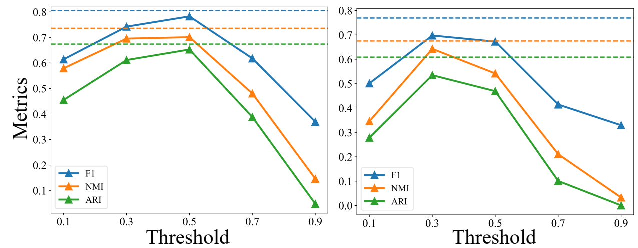

Figure 6. **The results of various anomaly detection (AD) methods under multiple fixed thresholds.** The horizontal dashed line represents the results of MEBin. Notably, the optimal fixed thresholds differ across AD methods, while our MEBin consistently outperforms the fixed threshold results.

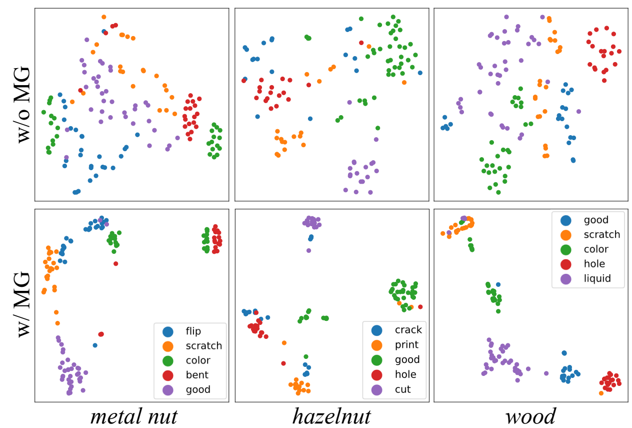

Figure 7. **T-SNE visualization of sub-images on the MVTec AD dataset.** We choose *metal nut*, *hazelnut* and *wood* as examples. The different colors of dots represent their anomaly classes.

[45], SimGCD [47] and AMEND [3]. As in our method, the AeBAD-S dataset is used as labeled abnormal images.

4.1.4. Evaluation metrics

We report three widely used metrics for the evaluation of clustering: the F_1 score, the normalized mutual information (NMI) [32], and the adjusted rand index (ARI) [36]. We use the Hungarian algorithm [27] to match the predicted clusters to the ground truth labels. For methods using anomaly maps, we measure the segmentation per-region overlap (AUPRO) with 30% FPR [5] to explore the impact of anomaly detection performance on the multi-class classification. All metrics are calculated by the official code.

4.2. Quantitative results

We report the results of the multi-class anomaly classification on the MVTec AD and MTD datasets in Table 1 and 2. Notably, AnomalyNCD is compatible with all the anomaly detection methods to enable subsequent multi-class anomaly classification.

All methods in Table 1 only use unlabeled images as input. Compared to Anomaly Clustering [40], AnomalyNCD combined with zero-shot anomaly detection method MuSc [30], achieves 8.8% gains on NMI, 9.5% gains on ARI, and 10.8% F_1 gains, proving that AnomalyNCD extracts more discriminative features than Anomaly Clustering through contrastive learning in anomalous regions. AnomalyNCD also outperforms state-of-the-art NCD methods, achieving 16.1% gains on NMI, 18.0% gains on ARI, and 14.3% F_1 gains. The performance advantage of AnomalyNCD is due to focusing on anomalies inside images.

In Table 2, all methods utilize both unlabeled images and labeled normal images from the same product. We integrate AnomalyNCD with various one-class AD methods. Utilizing CPR, which has the highest AUPRO, achieves the best performance on the MVTec AD dataset: NMI increases by 10.7%, ARI by 14.9%, and F_1 score by 9.6%, surpassing other methods. On the MTD dataset, although the segmentation AUPRO of all AD methods is lower than that of Uniformaly, PatchCore+AnomalyNCD still achieves improvements of 6.8% in ARI and 0.8% in F_1 score.

4.3. Ablation Study

4.3.1. Effectiveness of main element binarization

Table 3 compares MEBin's performance with the Otsu method [35] and fixed thresholds. Since the missed detection is more critical than over-detection in industrial scenarios, we define an anomaly as detected if the intersection over union (IoU) between the actual and predicted bounding boxes exceeds 0.1. Then we calculate the FPR and FNR for these binarization methods. MEBin demonstrates lower FPR and FNR than fixed thresholds and Otsu, leading to better multi-class classification results. Compared to the fixed thresholds, MEBin adaptively determines an optimal threshold for each image adaptation and achieves a 7.2% F_1 improvement. The Otsu tends to over-detect, particularly in normal images, making it less effective for this task.

In Fig. 6, we further apply MEBin on different one-class AD methods, which exhibit varying optimal fixed thresholds. For example, the optimal threshold of CPR [29] is near 0.5, whereas for PNI [2] it is approximately 0.3. MEBin's adaptive threshold selection for each image outperforms all fixed thresholds for the whole image set.

Mask Mechanism	NMI	ARI	F_1
(a) w/o MGA	0.598	0.494	0.698
(b) all tokens	0.507	0.382	0.600
(c) patch tokens	0.563	0.467	0.686
(d) class token (Ours)	**0.613**	**0.526**	**0.712**

Table 4. Compared with different mask attention mechanisms on the MVTec AD dataset, the best-performing result is in bold.

	MVTec AD			MTD		
L_m	NMI	ARI	F_1	NMI	ARI	F_1
1	0.606	0.508	0.690	0.191	0.157	0.420
3	0.608	0.511	0.694	0.209	0.173	0.436
6	**0.613**	0.519	**0.713**	0.253	0.214	0.492
9	**0.613**	**0.526**	0.712	**0.268**	**0.228**	**0.509**
12	0.609	0.521	0.712	0.249	0.213	0.492

Table 5. Ablation study on the position of mask-guided layers on MVTec AD and MTD datasets. Best results in bold.

4.3.2. Discussion of the mask-guided attention

The mask-guided attention (MGA) directs ViT to focus on tokens within the mask (anomalous region). We show the t-SNE visualization in Fig. 7. With MGA in the 2^{nd} row, image features have larger inter-class distances (e.g. *metal nut*) and smaller intra-class distances (e.g. *hazelnut* and *wood*) than those without MGA in the 1^{st} row. In terms of metrics on MVTec AD, our MGA has a 1.5% NMI gain, 3.2% ARI gain, and 1.4% F_1 gain shown in Table 4.

We conduct experiments to introduce the binary mask into different tokens in Table 4. (b) introduces the binary mask on `[CLS]` token and patch tokens. (c) introduces it on patch tokens. Our MGA (d) only performs on `[CLS]` token, which is directly input into the classifier. We make patch tokens have a global receptive field to introduce more surrounding information indirectly into `[CLS]` token. Compared to other mask strategies, our MGA achieves 5.0% NMI, 5.9% ARI, and 2.6% F_1 improvements.

4.3.3. The influence of the mask-guided layer L_m

In our method, we introduce mask-guided attention in the last L_m layers of ViT. As reported in Table 5, we conduct the experiments with different L_m. The results demonstrate that using mask-guided attention in the last 9 layers yields better performance on both datasets. When L_m is small, the `[CLS]` token cannot adequately focus on the anomaly within the binary mask. In contrast, when $L_m = 12$, the `[CLS]` token completely lost contextual information around the anomaly, resulting in a drop in metrics.

4.3.4. Discussion of merging strategies

In Table 6, we report results using different merging strategies. (i) averaging the predictions of the sub-images, (ii) utilizing the anomaly scores of the sub-images as weights to merge, and (iii) using our region merging strategy. Strategy (iii) achieves better metrics than (i) and (ii) on both datasets. In anomaly detection, noise inevitably leads to over-detections, which have small areas but large anomaly scores. Directly averaging results over all sub-images is inaccurate, especially when many false positive sub-images are involved. Similarly, using anomaly scores as weights is also inappropriate. In contrast, our merging method, which assigns weights to multiple sub-image predictions based on area, achieves more robust merged predictions.

	MVTec AD			MTD		
Merge	NMI	ARI	F_1	NMI	ARI	F_1
(i) Avg	0.610	0.521	0.709	0.257	0.223	0.501
(ii) Score Avg	0.600	0.513	0.703	0.228	0.208	0.485
(iii) Area Avg	**0.613**	**0.526**	**0.712**	**0.268**	**0.228**	**0.509**

Table 6. Compared with different merging strategies on the MVTec AD and MTD datasets.

	Precision	Recall	NMI	ARI	F_1
w/o PLC	0.797	0.727	0.597	0.517	**0.714**
w PLC	**0.821**	**0.876**	**0.613**	**0.526**	0.712

Table 7. The ablation experiment of pseudo label correction (PLC) on the MVTec AD dataset. We evaluate the precision and recall of the normal class.

4.3.5. Effect of Pseudo Label Correcting

Our Pseudo Label Correcting (PLC) aims to reduce the impact of over-detections from the anomaly detection process. Although over-detected sub-images are normal, they have large appearance differences, which makes it difficult to correctly group them into the same normal class. Table 7 presents the precision and recall of the normal class. After applying PLC, recall improves by 14.9%, which means that some misclassified normal sub-images are corrected by our PLC. Additionally, a 1.6% improvement in NMI is observed in the multi-class classification results.

5. Conclusion

In this paper, we propose a novel multi-class anomaly classification method AnomalyNCD, which is compatible with existing anomaly detection methods. Firstly, we propose the main element binarization approach to isolate most over-detections and missed detections brought by anomaly detection methods. Secondly, we crop each anomalous region and leverage mask-guided representation learning to focus the network on the local anomaly and learn discriminative representations. Finally, our region merging strategy achieves flexible classification at sub-image and image levels during inference. While the performance of AnomalyNCD is generally influenced by the quality of the anomaly detection method, our AnomalyNCD outperforms all the current industrial anomaly clustering methods and NCD methods in the nature scene.

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